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ORIGINAL ARTICLE

Floor plan creation using a low-cost 360° camera

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Abstract

The creation of a 2D floor plan is an integral part of finishing a building construction. Legal obligations in different states often include submitting a precise floor plan for ownership purposes, as the building needs to be divided between new residents with reasonable precision. Common practice for floor plan generation includes manual measurements (tape or laser) and laser scanning (static or SLAM). In this paper, a novel approach is proposed using spherical photogrammetry, which is becoming increasingly popular due to its versatility, low cost and unexplored possibilities. Workflow is also noticeably faster than other methods, as video acquisition is rapid, on a par with SLAM. The accuracy and reliability of the measurements are then experimentally verified, comparing the results with established methods.

KEYWORDS
floor plan, photogrammetry, point cloud, spherical, structure from motion

INTRODUCTION

Floor plans are an integral part of building documentation. A part of indoor modelling, they are part of building information modelling (BIM) expansion. 2D floor plans represent a simple way of visualising the disposition of rooms with dimensions and area. They find use in emergency room planning, facility management and, in the field of modern technology, in augmented reality and indoor navigation.

As of late, floor plan generation has been a subject of studies improving the data acquisition process and convenience. Manual measurement is subject to human error and is generally slow, inefficient, costly and potentially inaccurate for 2D purposes (in the case of non-orthogonal structures). A very precise instrument for measuring floor dimensions is a total station, which also bears the burden of slow speed. A static laser scanner is a better option, making data acquisition more automatic, but comes with the necessity of individual scan registration and is quite slow too.

One of the most convenient approaches is simultaneous localisation and mapping (SLAM). SLAM scanner operators are able to obtain point clouds with reasonable accuracy in a single pass. SLAM comes with some well-known limitations, as long scans without overlap can become shifted. The point cloud is subsequently sliced using computer-aided design (CAD) software and vectorised in 2D. Another modern approach is to model the point cloud in 3D for BIM purposes (Pavelka Jr. & Michalík, 2019; Pavelka et al., 2019), then generate a floor plan from this model.

As all the approaches mentioned above generally require expensive equipment, a photogrammetry-based solution is proposed, with a spherical camera as the instrument. Spherical photogrammetry is on the rise with increasingly reliable 360° cameras arriving on the consumer-grade market (Insta360, GoPro, Samsung among many others). Devices are often equipped with two fisheye cameras, but there can be numerous sensors resulting in better stitching (Teppati Losè et al., 2018). Their most useful feature is ensured overlap, because of the total field of view (Barazzetti et al., 2018). This drastically reduces the required time, images captured and overall requirements for successful alignment. Their capabilities in the photogrammetry field have been explored in many studies. Some of the possible uses are found in heritage documentation (Teppati Losè, Chiabrando et al., 2021), metric rectification (Barazzetti, 2022), street mapping (Barazzetti et al., 2022; Cao et al., 2022; Cera & Campi, 2022), agricultural (Castanheiro et al., 2020) and indoor modelling (Kwiatek & Tokarczyk, 2015; Teppati Losè, Chiabrando et al., 2021). In terms of accuracy, spherical cameras do not achieve the same level as traditional frame cameras, because of the smaller resolution per given field of view. They can be used to connect multiple bundles of regular photogrammetric data into one model with moderate success, albeit not with the accuracy of a total station (Barazzetti et al., 2019).

Enterprise-level spherical cameras offer higher resolution with the possibility of accurate prior calibration, resulting in a precise reconstruction (Zhao, 2021), but the cost is also noticeably higher. The resolution of consumer cameras may be improved in the future. Photogrammetry is inherently dependent on good light conditions, unlike laser scanning, where external light is necessary only to colour the cloud. Low-light performance can, however, be improved with large fisheye sensors, ensuring large aperture and high amount of light to the camera sensor.

Long lines of aligned spherical photos can, however, result in drastic shifts on the endpoints, even more notable than in the case of SLAM (Barazzetti et al., 2022; Teppati Losè, Chiabrando, et al., 2021). To mitigate this, it is advised to connect the lines with intersections or additional control points (CPs) to avoid error cumulation.

Some research has already been done on floor plan recovering using spherical cameras (Pintore et al., 2018). However, spherical images can also be processed in traditional photogrammetric software, using a standard structure-from-motion (SfM) pipeline (Pagani & Stricker, 2011). The industry standard software Agisoft Metashape 1.8.3 (build 14,298) (Agisoft Metashape Professional, 2022) was chosen, since it supports equirectangular projection used in 360° images (Cutugno et al., 2022). The result is a point cloud in the correct scale, which can be used in a similar manner as a terrestrial laser scanner cloud.

The resulting point cloud was compared with a SLAM cloud from a Leica BLK2GO scanner. BLK2GO is a handheld scanner with competitive parameters, simple workflow and lightweight construction. Not many studies have been conducted on this specific scanner since its arrival in 2019, but the overall impression is positive, whether in indoor (Piniotis et al., 2020) or outdoor (Del Duca & Machado, 2023) situations, where it becomes a trustworthy alternative to static scanners. Whereas SLAM clouds tend to be noisier, they offer better completeness and homogenous data than static scanners. BLK2GO also competes well with its predecessor/twin BLK360 (Dlesk et al., 2022). Other SLAM solutions have shown even better results (Pavelka et al., 2021). New possibilities have emerged with low-cost lidar sensors in iPhones and iPads, though their accuracy is insufficient and further development is needed (Teppati Losè et al., 2022).

The point cloud can be subsequently vectorised into a 2D floor plan. The simplest approach is to slice the cloud with a cross section and draw along in CAD software. Recent development is focused on extracting the floor plan automatically along with semantic information about walls, floors and ceilings. The Hough transform can be used to automatically draw a 2D floor plan (Okorn et al., 2010). Some studies used extrusion (Stojanovic et al., 2019) or other algorithms (such as RANSAC) to extract a full 3D floor plan from the point cloud (Hossein Pouraghdam et al., 2019). Deep learning techniques with stacked neural networks are also trending with their ability to segment and locate elements on raster (Mask R-CNN, YOLO, SSD) and point cloud data (ConvPoint, PointNet) with some applications in outdoor/indoor modelling (Cao et al., 2022; Karara et al., 2021). These techniques may find use in detecting objects such as doors and windows, integrating them directly into a floor plan.

We present our case study on a construction site of a five-storey building with hundreds of rooms. Consistent with the idea of a rapid and convenient acquisition, only a few ground control points (GCPs) outside of the structure were used to obtain a well-scaled model, with no CPs inside the building. As there is no requirement for absolute position in the coordinate system and we are concerned primarily about relative dimensions of the structure, only numerous indoor length dimensions were measured to compare the model with and observe the propagation of alignment errors throughout the building. Photos from an unmanned aerial vehicle (UAV) were also used to connect different branches of spherical positions and firm the alignment from the outside. Some studies already explored such combinations of datasets, using aerial images as a rigid block to orient spherical data (Teppati Losè, Spanò et al., 2021).

Our objective is to determine whether spherical photogrammetry can achieve sufficient accuracy compared to the verified SLAM approach. The objective is to find a suitable replacement at a fraction of the cost of SLAM, which can also be comparably fast and simple, hence only six CPs in the exterior were placed. Using more CPs inside the building would undeniably result in better accuracy, but it would disregard our premise.

MATERIALS AND METHODS

Facility overview

The aforementioned construction site is located in the western part of Czech Republic, near the spa town of Karlovy Vary. It is to be a five-storey residential building with over forty apartments, each with three to five rooms. It consists of two intertwined parts connected via staircases with alternating "half-storeys" as seen in Figure 1. The longest diagonal is over 60 m long.

At this stage of the construction, interior spaces are heavily in progress. Rooms sometimes contain some clutter, although the walls are most of the time visible. The construction is concrete-based and the walls consist of red hollow bricks. This is important, as this surface has a rough texture, providing enough data for scale-invariant feature transform (SIFT) photogrammetry algorithms to detect tie points for image alignment. Some of the floors already have inner partition walls, some are yet to be constructed. Unfortunately, the places with inner partitions are sometimes quite dark, raising concerns about the viability of our method. The lowest (basement) floor was omitted from comparison with SLAM due to very bad lighting conditions.

Instruments used

For spherical image acquisition, the Insta360 One RS Twin Edition (Figure 2) (Insta360 One RS, n.d.) was used. The 360° module is equipped with two 1/2″ fisheye sensors, each with a field of view over 180°, providing sufficient overlap for image stitching. The pixel size is 1.2 μm, lenses have 7.2 mm equivalent focal length and f/2.0. The resulting video is 5760 × 2880 pixels.

FIGURE 1 Overview of the site from above.

FIGURE 2 Insta360 One RS Twin Edition.

Insta360 spherical cameras save the dual-fisheye pictures in .insp/.insv format, each camera to one file. Aside from image data, it also contains metadata including gyro and position information from a Global Navigation Satellite System (GNSS) capable connected mobile device. These photos and videos can be imported to the Insta360 Studio desktop application for stitching, colour correction, stabilisation and other. The result is footage in 2:1 equirectangular projection. After exporting stitched footage in .mp4, image frames from the video can be extracted in various software, for example, Blender (Blender Online Community, 2018). The frames can be extracted from a regular video too, although timelapse mode is more practical due to the much smaller video size. The timelapse video mode was chosen, as it is the fastest way of acquisition. The camera also offers interval mode, capturing photos, but the fastest interval possible is 3 s, which is not suitable for SfM purposes (Teppati Losè, Chiabrando et al., 2021). To ensure enough data, half-second intervals (or two frames per second) were determined as a reasonable compromise.

Using frames extracted from a video for photogrammetry purposes (also called videogrammetry; Brilakis et al., 2011; Gruen, 1997) is a known way of simplifying the acquisition process with satisfactory results (Torresani & Remondino, 2019). Compression plays a significant role in alignment accuracy and reconstruction quality. While lossless formats such as RAW or tiff produce the best quality, using a reasonably compressed .jpg file achieves comparable results (Alfio et al., 2020).

It is debatable whether to use stitched equirectangular panoramas (Figure 3) or single photos from each sensor, as the stitching process is a “black box” in proprietary software. Interior orientation parameters of stitched spherical images cannot be characterised or adjusted due to discontinuity along the stitching line(s).

FIGURE 3 Equirectangular projection of a spherical photo with interior typical for this site.

FIGURE 4 DJI Mini 2.

Some studies suggest better resulting accuracy with single photos, at the cost of longer processing time and more complicated workflow. The two sensors may have different projection centres and processing them unstitched also allows Metashape to use a fisheye camera distortion model, better estimating their interior orientation parameters, whereas the spherical model does not take any distortion into account (Agisoft Metashape Professional, 2022) and the internal orientation model only contains focal distance. Processing single images in master–slave fashion with known relation between cameras might result in more accurate, but also incomplete alignment (Teppati Losè, Chiabrando et al., 2021). Nevertheless, the spherical camera approach was chosen, as the dataset was going to be large enough already and additional workload might increase processing time substantially.

The UAV used to capture images from above is the DJI Mini 2 (Figure 4; DJI Mini 2, n.d.). This entry-level UAV offers a 12 MPx camera with 1/2.3″ sensor, 83° field of view and 24 mm equivalent focal length.

The point cloud to be used as ground truth was acquired using a Leica BLK2GO (Figure 5; Leica BLK2GO, n.d.). This laser scanner uses visual and lidar SLAM algorithms to estimate its position while capturing up to 420,000 points per second. The manufacturer states the scanner achieves ±10 mm accuracy indoors with noise ranging up to 3 mm. The scanner operates in a 25 m range and weighs 775 g.

FIGURE 5 Leica BLK2GO.

Acquisition

Photogrammetry data

The Insta360 One RS was mounted on a metre-long pole. The pole was held above the operator's head during the facility walkthrough so that his presence in images was minimised. Due to varying light conditions on the site, automatic exposure was set. Some studies mention using preliminary GPS coordinates to ensure and accelerate successful alignment (Barazzetti et al., 2022). However, this method cannot be applied in closed environments such as this.

In order to improve the accuracy of alignment, it is recommended to avoid dead ends or long lines without any checks, and secure overlaps and crossings between paths. The building includes many openings to the outside through unfinished windows and balconies. To avoid model deformations, the inside and outside were connected using a UAV and leaning out with the spherical camera. While the UAV may not be crucial to the alignment, it was employed to ensure sufficient connection across the photogrammetric model. Almost every possible opening along the building periphery was exploited to connect the model with the outside. There were a few rooms with finished windows that could not be opened, so the alignment was not connected; these premises were aligned with severely worse precision.

The entire building was captured with the 360° camera except for the lowest floor, which was very dark for photogrammetry purposes. Every room was captured with one walkthrough from multiple sides. In the end, 7262 spherical images were captured. As the videos were framed every 0.5 s, the acquisition time was a little over 1 h.

The UAV was equipped with a blade protector in case of a collision in the busy construction environment. It was flown manually all around the building, maintaining around 10 m distance and sufficient overlap with enough footage of the CPs placed around the facility. Footage contained both oblique and nadir photos, with part of the dataset taken perpendicular to outer walls with attention to the openings. Ninety-one aerial JPEG photos were captured.

Laser scanning

To compare photogrammetry results, the SLAM scanner was used on the first floor above ground level. The acquisition was conducted in a similar manner as with the 360° camera, with analogous routes and speed. The whole double-floor was captured in two separate walkthroughs with an overlap, first the southern part, then the northern part. The two resulting point clouds were subsequently aligned and exported in .e57 format.

Photogrammetry processing

Alignment

All images were imported to Agisoft Metashape. Frames from the 360° camera were set as "spherical", UAV images remained as "frames". The images were aligned on "high" accuracy with generic preselection to speed up the complicated alignment. As there was no preliminary information about the position of the cameras, reference preselection was disabled and sequential preselection would result in error cumulation. Alignment took over 15 h and was successful, with 7341 out of 7352 cameras aligned. To optimise the alignment, the sparse cloud was reduced from 7.4 million points to 4 million, omitting points with subpar accuracy metrics. The final mean reprojection error was 3.2 pixels.

To scale the model appropriately, six GCPs were placed on the external corners of the facility, measured with a total station and a prism on a pole. After alignment they were marked on a sufficient amount of both spherical and aerial photos and "Optimise cameras" was used to optimise the alignment. The prior accuracy of a GCP was set to 1 cm. To better estimate the scaling accuracy, two points (3 and 4) were set apart as CPs and recalculated. The calculated residuals on the control and check points can be found in Table 1.

Dense reconstruction

After successful alignment with verified accuracy, the dense point cloud was computed. This process is especially time-consuming and computationally intensive. The initial attempt was to calculate the cloud with a "high" accuracy setting, which downscales the images by a 2×2 factor and generates depth maps from which Metashape derives the point cloud. However, the final stage proved to be too difficult for our computer, even with its competitive hardware (NVIDIA RTX 3060, AMD 5900X, 32 GB RAM), as in the final stage the computer ran out of memory.

As a substitute, it was decided to change the workflow and calculate the mesh from the depth maps instead, and subsequently sample points on the mesh every 5 cm. This proved to be a much faster method, one that also interpolates the cloud, filtering outliers. However, it also filters potentially desirable points in darker areas.

After the workstation was upgraded with an additional 32 GB of RAM (totalling 64 GB), another attempt at dense cloud reconstruction was successful, resulting in over 1 billion points. The two mentioned results are compared further.

TABLE 1 Residuals on GCPs and CPs.

GCP	*X* error (cm)	*Y* error (cm)	*Z* error (cm)	Total error (cm)	Image (Px)
1	−0.65	−0.40	4.12	4.19	0.272
2	−1.73	3.72	0.85	4.19	0.144
5	3.77	0.87	1.15	4.04	0.153
6	−2.16	−3.09	−2.14	4.34	0.559
Total	**2.78**	**1.56**	**2.43**	**4.19**	**0.166**
CP	***X* error (cm)**	***Y* error (cm)**	***Z* error (cm)**	**Total error (cm)**	**Image (Px)**
3	3.76	−3.69	−2.21	5.71	0.152
4	−2.51	1.86	1.85	3.63	0.144
Total	**3.20**	**2.92**	**2.03**	**4.78**	**0.148**

Floor plan generation

The point cloud (either SLAM or photogrammetric) was exported to an interchangeable format and imported to Autodesk ReCap, where the cloud was cleaned and unnecessary points were removed. The resulting .rcp file can be attached in AutoCAD, sliced with a cross section and vectorised in top view. The cross section can have variable thickness depending at which height the floor plan is to be drawn. With regards to noise occurring along the walls, the cross sections were chosen to be as thick as possible to average the noise with the line drawing.

RESULTS

In this section, various methods are used to determine the validity of the spherical photogrammetric method. As a quantitative comparison, metrics of point-to-point distance of the cloud are calculated. The final results of measured floor plan areas are also compared. For qualitative comparison we look at quality of photogrammetric reconstruction, namely in dark areas, and find other sources of error, for example, potential shifts of the alignment and their impact.

Quantitative analysis

Cloud-to-cloud metrics

Table 2 shows processing times and attributes of the acquired point clouds. SLAM and photogrammetry (DENSE) clouds are similarly dense, while the sampled cloud (MESH) is much sparser due to 5 cm sampling. This sparsity comes in handy when using the cloud for vectorisation as it already contains enough information. The two photogrammetry clouds are computed from the same depth maps, which took exactly 5 h to process (included in the processing time column). As Table 2 shows, the processing time is reduced substantially with the mesh created directly from depth maps. Point cloud sampling is then done in a matter of minutes. The listed number of points is after limiting clouds to the compared (lowest) floor. The unsegmented dense cloud amounted to over 1 billion points; for further uses it was subsampled to 1 cm density.

To perform cloud-to-cloud analysis, the comparison was limited to the first floor above ground level. The resulting point clouds were cleaned and imported to the CloudCompare software (CloudCompare, 2022). The two clouds resulting from photogrammetric reconstruction were already aligned. The SLAM cloud was roughly aligned to the others with eight CPs using similarity transformation (without scale adjustment). Iterative precise alignment with RMS minimisation was then performed by CloudCompare itself. To make the comparison more objective, the cloud was rid of all the parts that exceeded the extent of the SLAM cloud (ground truth) such as outside walls, not measured by SLAM, as the main focus was indoors. As the data acquisition took place on different days, the SLAM and photogrammetric data differ slightly due to moved construction equipment and material on the site. These differences were also partly mitigated by excluding the parts from comparison. Also, some areas were not

TABLE 2 Parameters of compared point clouds.

Type of cloud	Number of points	Alignment time	Processing time
SLAM	126,796,266	—	—
DENSE	97,627,193	15 h 52 min	17 h 20 min
MESH	1,497,097	15 h 52 min	7 h 28 min

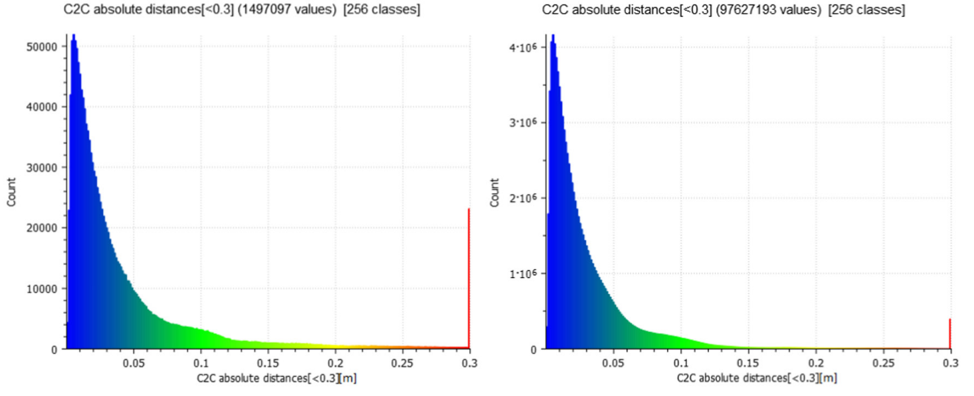

FIGURE 6 Histograms of cloud-to-cloud comparisons. DENSE cloud (left) and MESH cloud (right).

scanned thoroughly with the BLK2GO scanner, resulting in a sparser cloud, thus increasing the cloud-to-cloud metric for these areas.

Figure 6 shows two histograms of calculated cloud-to-cloud distances from CloudCompare. The SLAM cloud was set as ground truth and photogrammetric clouds were compared. As expected, the histograms show a similar curve, with the DENSE cloud achieving smaller errors than MESH. The *x* axis of the graph only shows 0–30 cm distances, with bigger distances pooled in the 30 cm bracket. The MESH cloud contains much more of these grave errors mostly due to the aforementioned interpolation in very sparse areas.

Cloud-to-cloud comparison only measures absolute distances, without taking in account the direction of the errors. These results do not follow Gaussian distribution, so they cannot be fitted, and the standard deviation cannot be computed. To obtain a symmetrical histogram with negative and positive errors, it is necessary to know the direction of the error. One such method is to compute the mesh and calculate signed cloud-to-mesh distances. The SLAM cloud was therefore meshed in Metashape (with interpolation) and imported back to CloudCompare, where new histograms were calculated (see Figure 7). Another possible approach is to compute M3C2 distance (Lague et al., 2013) using a dedicated CloudCompare plug-in, which computes the local distance between two point clouds along the normal surface direction which tracks 3D variations in surface orientation.

To estimate the statistical parameters of the curves, the histograms are fitted with normal distribution. When fitting the original histograms, which contain many outliers, standard deviation is very high and does not fit the bell curve of the histograms. With outliers cut to ensure the best fit of the curve, characterising random error (noise), standard deviation results are much lower. The threshold of outliers beyond the normal distribution was set to 6 cm to ensure a good fit of the Gauss distribution. Outliers may be caused by large systematic shifts, changed objects between measurements and false interpolations of the MESH cloud. The statistics are listed in Table 3.

Comparison of area measurements

The next comparison is aimed to find differences in the final product—the areas of the rooms calculated from the floor plan. Rooms are numbered 1–9 (Figure 8), and they were vectorised from two point clouds: MESH and SLAM. The MESH cloud (along with the DENSE cloud) is incomplete in dark places and the exact wall lines had to be partially guessed sometimes.

The results are shown in Table 4. Room no. 7 is the one with the biggest observed shift, depicted in Figure 10. Here, it resulted in over 5 m^2 difference, which is explained further in the “Shifts” section. Other rooms are constructed with about 0.5 m^2 accuracy. Without the one clear outlier, the resulting standard error is 0.49 m^2.

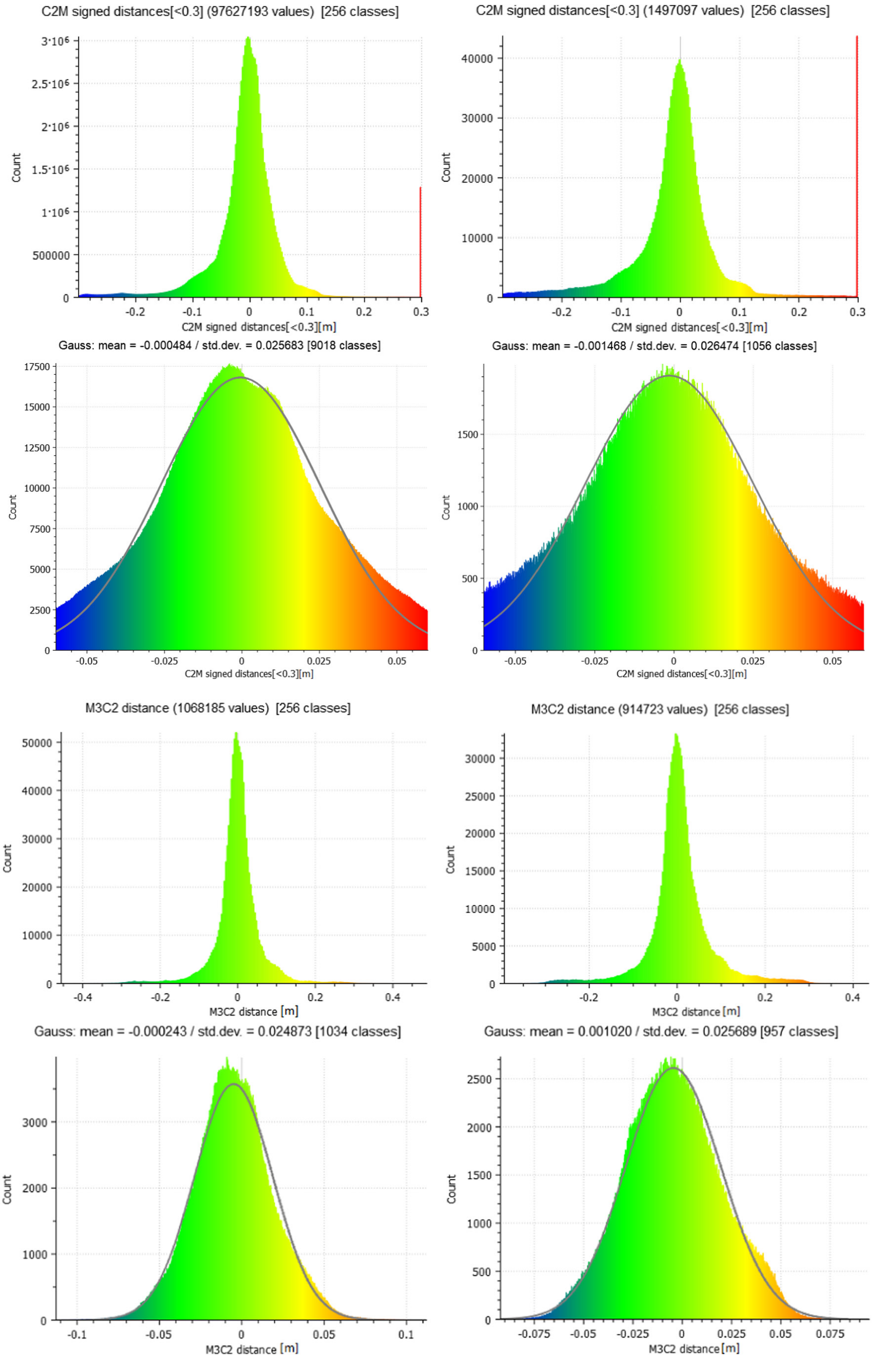

FIGURE 7 Histograms of cloud distance comparisons. DENSE cloud (left) and MESH cloud (right). The first two rows of images depict cloud-to-mesh distances, the last two rows M3C2 distances. The second and fourth rows show histograms without outliers, fitted with normal distribution.

TABLE 3 Statistical parameters of cloud-to-mesh distances. Outliers were cut above 6 cm absolute distance.

	Original histograms		Without outliers	
Type of cloud	**Mean (cm)**	**Std. dev. (cm)**	**Mean (cm)**	**Std. dev. (cm)**
DENSE	−0.50	6.41	−0.05	2.57
MESH	−0.49	8.69	−0.15	2.65
DENSE (M3C2)	−0.03	5.62	−0.02	2.49
MESH (M3C2)	0.05	6.99	0.1	2.57

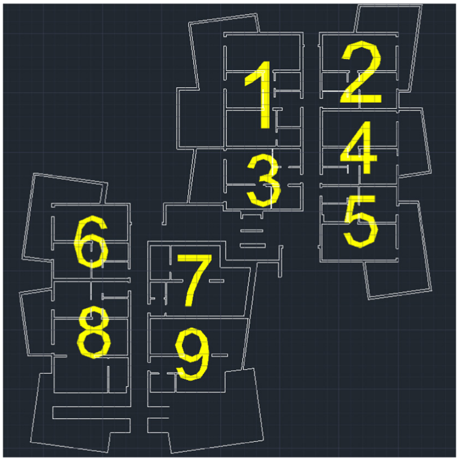

FIGURE 8 Floor plan with numbered rooms.

TABLE 4 Areas resulting from vectorisation.

Room no.	MESH (m^2)	SLAM (m^2)	Difference (m^2)	Difference (%)
1	87.27	87.77	0.50	0.57
2	57.85	58.45	0.60	1.03
3	48.07	48.18	0.11	0.23
4	58.25	58.34	0.09	0.15
5	58.41	58.41	0.00	0.00
6	57.66	58.54	0.88	1.50
7	**68.12**	**73.64**	**5.52**	**7.50**
8	87.75	88.18	0.43	0.49
9	70.84	70.29	−0.55	−0.78

They are stressed, because they represent a significant outlier, described further in the “Shifts” chapter.

Qualitative analysis

Completeness

In dark places, especially the cellar cubicles, the photogrammetric reconstruction was fatally incomplete. However, that is not a surprise, and missing dimensions had to be determined by manual tape measurement. The point cloud sometimes calculates some visible points that can be used to draw the floor plan, but they are often insufficient. The mesh-based sampled cloud shows attempts at interpolating the very sparse depth map information, which results in a curved and deformed outline of the walls and does not bring any improvement (see Figure 9 comparing cross sections of MESH and SLAM clouds). It is clear that good light conditions are crucial for SfM processing.

Shifts

In rooms with barred windows, without the possibility to connect the model with the outside, the model is often visibly shifted. One particular case is visible in Figure 10 (room no. 7), where the shift amounts up to 40 cm in the right wall. As the model of the room is connected with the rest only through one door, often poorly lit, it does not come as a surprise.

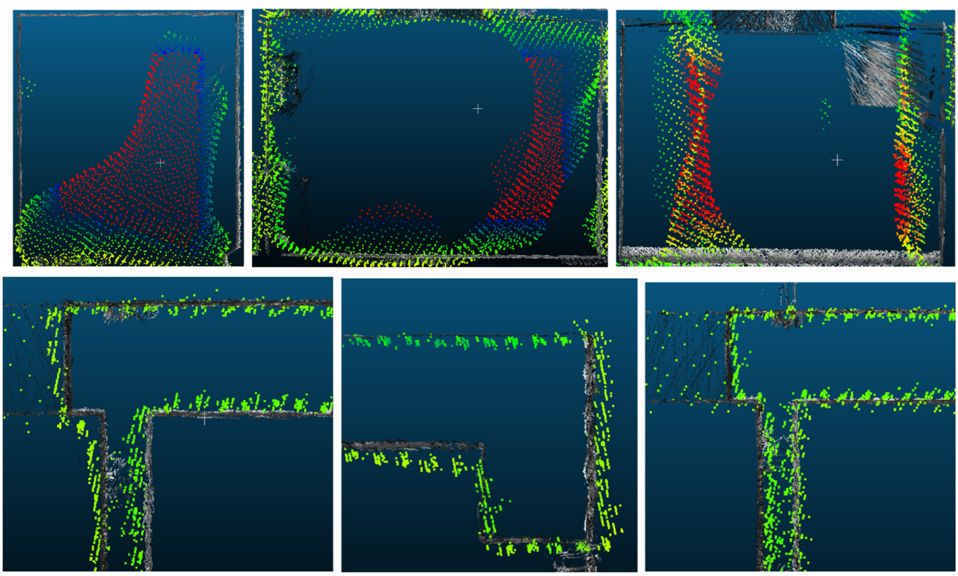

FIGURE 9 Examples of failed interpolation of meshing in dark areas (first row). High errors are marked in red. Examples of successful reconstruction in good light conditions are in the second row.

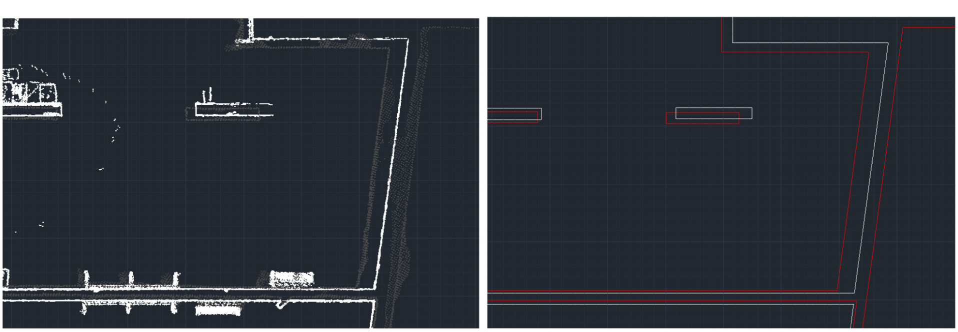

FIGURE 10 A significant shift of the model observed in room no. 7 on cross sections of point clouds (SLAM white, MESH grey) and vectorised floor plans (SLAM white, MESH red).

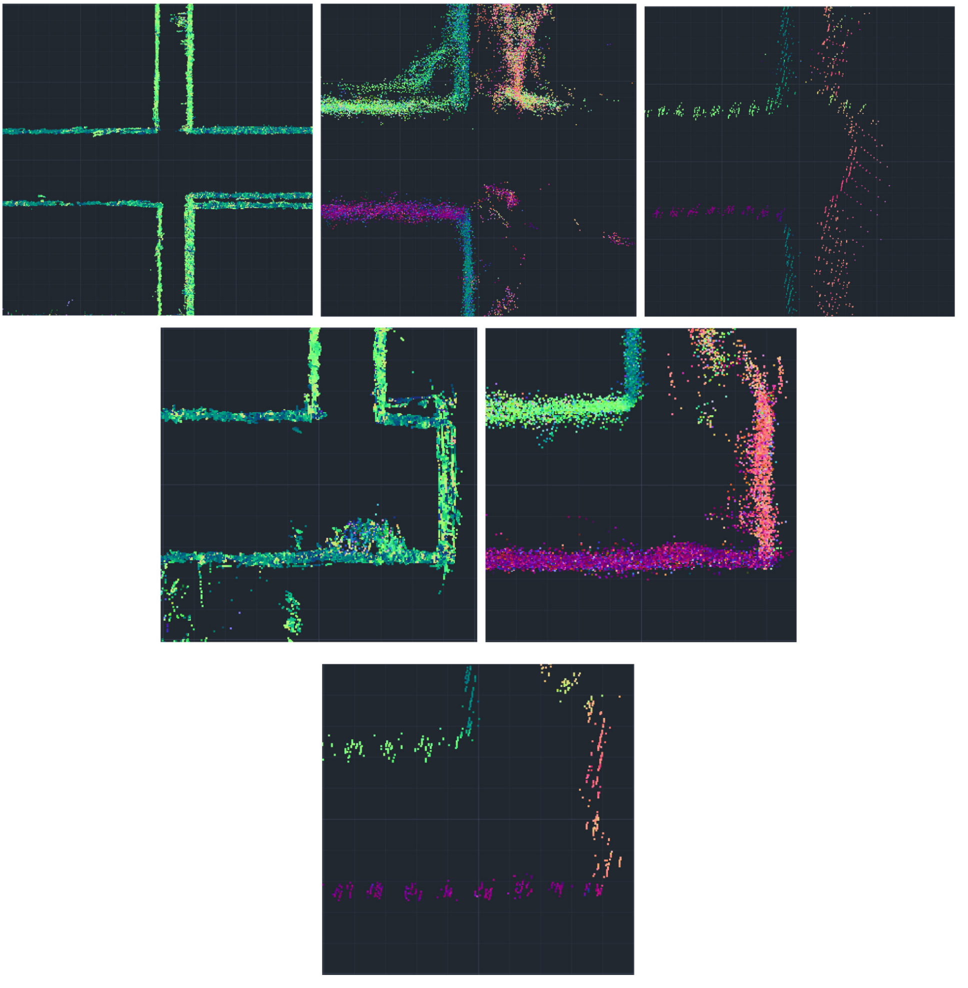

FIGURE 11 Two-metre thick cross sections of the clouds SLAM (left), DENSE (middle) and MESH (right). All images in a row show the same area, the second row is in a higher detail. All clouds visualised using their normals.

The shifting is noticeable in a few other locations, although the magnitude of the shifts is not as big. Sometimes the walls are modelled thinner/thicker than in reality, again because of uncertain alignment of images through a doorway. This effect might be partially mitigated by denser sampling of videos, thus using more images for alignment at the expense of bigger time and computing requirements.

Noise in vectorisation

When drawing along the point cloud cross section, the points are spread across the wall, which affects the accuracy of the final drawing, depending on where the line is drawn. The walls consist of red bricks with regular alternating indentations around 1 cm deep. This also creates uncertainty along with the noise from SLAM and photogrammetry. The floor plan can be drawn at various heights, depending on whether it is desirable to add windows and other elements. The cross-section slice can be variably thick and with an accurate point cloud, there is no need to stretch it more than a decimetre. However, to smooth out noise present in the clouds, it was set around 2 m thick, almost the whole height of the room. Figure 11 shows various areas of vectorisation of all given clouds.

As illustrated, the SLAM cloud shows the smallest amount of noise, the DENSE cloud is significantly noisy and the MESH cloud, being created from an interpolated mesh, is less randomly noisy, but with similar uncertainty for vectorisation as the DENSE cloud. Noise in photogrammetric clouds is between 3 and 6 cm with the MESH cloud having less outliers, and the SLAM cloud is around 1–2 cm. The middle image of the first row also shows differences in quality in different light conditions, the lower right of the image being reconstructed in a darker environment, where the right wall is completely missing. The right image shows the MESH interpolation of such an uncertain area.

In the first row, photogrammetric clouds also increased the thickness of the adjacent wall. This is most likely due to small shifts that are not as drastic as in the case of room no. 7, but nevertheless can have an effect on the magnitude of centimetres. As discussed earlier, these happen because of suboptimal connection with other parts of the model.

The SLAM cloud is undoubtedly the most reliable. However, in the screenshots shown, it also displays a different type of error, where some walls are doubled. This is apparently caused by multiple passings with the scanner and their imperfect alignment.

DISCUSSION AND CONCLUSIONS

This research presented a novel approach of floor-plan creation and indoor modelling based on spherical photogrammetry. Its most valued attributes are speed and cheapness. The data acquisition phase is very similar to SLAM in both time and workflow. While the processing time took over 20 h in this case, it may be reduced with better computing hardware and optimised in further research with resolution and sampling frequency being some of the variables to fine-tune. Processing difficulty increases with size and complexity of the reconstructed building. The error, propagated to the interior, would also probably get larger. For smaller structures, the spherical photogrammetry approach is viable and effective.

Leica BLK2GO, and other SLAM scanners, offer a convenient and somewhat reliable way of providing indoor documentation for various purposes. However, its cost is discouraging, selling at around the 50,000-dollar price point. On the other hand, the Insta360 One RS costs around $550, a staggering difference. While the photogrammetric post-processing is more complicated and requires an advanced computer for reasonable results, the acquisition can be just as fast. Compared to any manual method, SLAM and photogrammetry are clearly faster and obtain more complex solutions.

One of the most constraining barriers in using spherical photogrammetry for indoor reconstruction is low-light situations. Many premises could not be reconstructed properly since the camera was not able to obtain enough information in darker areas. Because the room lights are not yet installed at this stage of construction, a possible solution to this problem is to place auxiliary battery-powered lights in low-lit areas. Another experimental approach is to place a 360° light coaxially just above/below the spherical camera, moving through the site with it. This would probably result in unevenly lit areas with shadows, which could make alignment more difficult. A partial mitigation is to use a camera with a bigger sensor and aperture, resulting in better low-light performance, such as the new Insta360 One RS 1-inch edition.

Another worrying aspect is the possible shifting of photogrammetric models, which can move rooms relative to each other by a significant magnitude. This can be mitigated by connecting the model with the outside through openings whenever possible.

As shown in the tables, the achievable accuracy of a vectorised wall can be a few centimetres, while the area accuracy is in tenths of m^2.

To accelerate the reconstruction process and to smooth out noise, direct meshing from calculated depth maps has shown some promise. The caveat is an even worse result in poorly lit areas.

Further research can concentrate on optimising and fine-tuning the acquisition and reconstruction parameters to obtain good quality in a reasonable amount of time. These parameters include sampling time of the captured spherical video frames, quality of dense reconstruction (down-sampling of the footage) or various interpolation algorithms.

Aside from the point cloud, another possible output of SfM processing would be an orthomosaic of a floor or wall. To achieve it in a reasonable amount of time, it is necessary to limit processed footage to a reasonable amount, as orthorectifying is a computationally intensive process. In the case of a floor, another caveat is the operator usually covering the lower part of frames, obscuring much of the floor. It may be possible to mask out the bottom image part, but the orthomosaic would be created from suboptimal angles.

Contemporary and future efforts are concentrated on automatic extraction of documentation, whether a simple floor plan or even a full BIM model, further eliminating the necessity of manual labour, prone to human errors.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

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